

1.1.7 Basic Logic Gates (AND, OR, NOT)

لاچک گیٹس الیکٹرانک سرکٹ ہیں جو بولین آپریشن انجام دیتے ہیں اور تمام ڈیجیٹل نظاموں کی بنیادی اینٹیں ہیں۔

CS XI

Chapter 1: Computer Systems

Lecture 04 of 12

★ Golden

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LEARN IT

Logic gates are electronic circuits that perform Boolean operations. They take one or more input bits and produce one output bit. They are the **building blocks of all digital systems** — a processor contains billions of them.

The three basic gates match the three basic operations: the **AND gate**, the **OR gate** and the **NOT gate** (also called an **inverter**).



$$\text{AND: } Y = A \cdot B$$



$$\text{OR: } Y = A + B$$



$$\text{NOT: } Y = A'$$

Symbols of the basic gates. AND has a flat back and round front ("D" shape); OR has a curved back and pointed front; NOT is a triangle with a small circle (bubble).

AND gate — $Y = A \cdot B$

A	B	Y
0	0	0
0	1	0
1	0	0
1	1	1

OR gate — $Y = A + B$

A	B	Y
0	0	0
0	1	1
1	0	1
1	1	1

NOT gate — $Y = A'$

A	Y
0	1
1	0

WORKED EXAMPLE

A school bell circuit. The bell (Y) should ring when the timer says "period over" ($A = 1$) **and** the school is open ($B = 1$).

Step 1: The bell needs both conditions, so use an AND gate: $Y = A \cdot B$.

Step 2: Sunday, timer fires: $A = 1, B = 0 \rightarrow Y = 1 \cdot 0 = 0$. No bell.

Step 3: Monday, timer fires: $A = 1, B = 1 \rightarrow Y = 1$. The bell rings.

If instead the bell should ring when *either* the timer fires *or* the principal presses a button, you would use an OR gate.

REMEMBER IT

Shortcut: AND output is 0 if *any* input is 0. OR output is 1 if *any* input is 1. Learn just the "special row": AND is 1 only at 11; OR is 0 only at 00.

COMMON MISTAKE

Do not mix up the AND and OR symbols. AND has a **flat** input side (like the letter D). OR has a **curved** input side and a **pointed** output, like a shield or arrowhead.

اردو میں سمجھیں

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EXAM TIP

You are often asked to "draw the symbol, write the expression and truth table" of a gate (2–3 marks each). Draw neat symbols with labelled inputs A, B and output Y. Write the full truth table in binary order. A frequent MCQ: "Which gate gives 1 only when all inputs are 1?" (AND).

CHECK YOURSELF

1. What is the output of an OR gate when $A = 0$ and $B = 0$?
2. What is another name for the NOT gate?
3. For which input does a 2-input AND gate output 1?

Answers: 1. 0. 2. Inverter. 3. Only $A = 1, B = 1$.

Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

MCQ 1. Logic gates implement:

- a. Boolean operations in hardware
- b. SQL joins
- c. HTTP only
- d. Pie charts

MCQ 2. AND output is 1 for inputs 1,1:

- a. 1
- b. 0
- c. Z
- d. X

MCQ 3. OR of 1 and 0 is:

- a. 1
- b. 0
- c. NAND
- d. XOR

MCQ 4. The inverter is the:

- a. NOT gate
- b. AND gate
- c. Bus
- d. Router

MCQ 5. Building blocks of digital systems are:

- a. Logic gates
- b. Spreadsheets
- c. MVPs
- d. Fonts

MCQ 6. $Y = A \cdot B$ is the:

- a. AND expression
- b. OR expression
- c. XOR
- d. NOR

Extra short questions

1. Define a logic gate.
2. State the AND, OR, NOT expressions.
3. Draw the truth table of 2-input AND from memory.
4. Why is NOT drawn with a bubble?

Extra long question

1. For AND, OR and NOT: symbol, expression, full truth table, and one daily-life model. ★ Golden.

Extra answer key

MCQs: 1. A 2. A 3. A 4. A 5. A 6. A

Short

1. An electronic circuit that performs a Boolean operation; the building block of digital systems.
2. $Y = A \cdot B$; $Y = A + B$; $Y = A'$ (or $\bar{A}$).
3. $00 \rightarrow 0$, $01 \rightarrow 0$, $10 \rightarrow 0$, $11 \rightarrow 1$.
4. The bubble marks inversion of the input.

Long

1. Standard IEC/US symbols. AND two locks, OR two doors, NOT a switch. Tables must have every row.

Next lecture: 1.1.7 Universal & Advanced Logic Gates